

Some notes on the trigonometric R method.

A. Introduction and proof. The trigonometric R -method is a technique that converts the sum of a sine and cosine into one term, i.e.

$$A \sin(x) + B \cos(x) = R \sin(x + C). \quad (1)$$

In these notes we will use capital letters ($A, B, C, \dots$) to represent constants, and x to be the variable of choice.

The technique is most commonly taught using the compound angle formula

$$\sin(x + C) = \sin(x) \cos(C) + \cos(x) \sin(C).$$

If we take this and compare the coefficients in Equation (1), we have

$$\begin{aligned} R \sin(x + C) &= R \cos(C) \cdot \sin(x) + R \sin(C) \cdot \cos(x) \\ &= A \sin(x) + B \cos(x), \end{aligned}$$

meaning

$$A = R \cos(C), \quad B = R \sin(C).$$

This is a set of simultaneous equations that we can solve. We assume $A, B > 0$ (if not, you need to be more careful). First, notice that

$$R = \frac{A}{\cos(C)} \quad \text{and} \quad R = \frac{B}{\sin(C)},$$

so

$$\begin{aligned} \frac{A}{\cos(C)} &= \frac{B}{\sin(C)} \implies \tan(C) = \frac{B}{A} \\ &\implies C = \arctan\left(\frac{B}{A}\right). \end{aligned}$$

Furthermore, if we square both of the simultaneous equations ($A^2 = R^2 \cos^2(C)$ and $B^2 = R^2 \sin^2(C)$) and add them together, we have

$$\begin{aligned} R^2 \cos^2(C) + R^2 \sin^2(C) &= A^2 + B^2 \implies R^2 = \frac{A^2 + B^2}{\sin^2(C) + \cos^2(C)} = A^2 + B^2 \\ &\implies R = \sqrt{A^2 + B^2} \end{aligned}$$

because $\sin^2(C) + \cos^2(C) = 1$ and $R \geq 0$ is the ‘amplitude’.

Remembering these two facts about C and R is technically enough to apply the method in practice. It is summarised in Equation (2) below.

B. On the alternative forms. You will often be asked to use the R -method to write two terms as a single cosine, e.g. write $A \sin(x) + B \cos(x) = R \cos(x + D)$ where D is

another constant. Instead of remembering the various forms of the R -method, I'd stick with just knowing the one variant

$$A \sin(x) + B \cos(x) = R \sin(x + C), \quad C = \arctan\left(\frac{B}{A}\right), \quad R = \sqrt{A^2 + B^2} \quad (2)$$

and instead use the fact that **cosine is the ‘cofunction’ of sine**. This means that

$$\sin(x) = \cos(90 - x) \quad (3)$$

for all x . Indeed, this identity holds for

$$\tan(x) = \cot(90 - x), \quad \sec(x) = \operatorname{cosec}(90 - x),$$

and the ‘co’ in each name is indicative that it is the so-called cofunction of the other (cos is the cofunction of sine, cot is the cofunction of the tangent, and cosec is the cofunction of the secant). Applying the cofunction identity in Equation (3)

$$\begin{aligned} A \sin(x) + B \cos(x) &= R \sin(x + C) \\ &= R \cos(90 - (x + C)) \\ &= R \cos(-x - C + 90). \end{aligned}$$

Now we use the fact the cosine is an even function, meaning $\cos(x) = \cos(-x)$, so continuing on,

$$A \sin(x) + B \cos(x) = \dots = R \cos(x + C - 90),$$

that is, $D = C - 90$.

Both of these facts (cofunctions and cosine being even) are very helpful tools to have, so use the trigonometric R -method as an excuse to help remember them!

C. A worked example. Suppose we are given

$$12 \sin(x) + 5 \cos(x)$$

and want to write this as just one term. Then, using Equation (2), we have $R = 13$ and $C = \arctan(5/12) = 22.62$ (2dp). Therefore

$$12 \sin(x) + 5 \cos(x) = 13 \sin(x + 22.62),$$

or alternatively, with $22.62 - 90 = -67.38$,

$$12 \sin(x) + 5 \cos(x) = 13 \cos(x - 67.38).$$

You can always check the R method by plugging in any values of x into your calculator.